

AMBERLEY, Australia

WMO: 945680

Lat: 27.6297S

Lon: 152.7111E

Elev: 25

StdP: 101.02

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 7	(b) 1.0	(c) 2.4	(d) -6.8	(e) 2.1	(f) 21.5	(g) -4.0	(h) 2.7	(i) 17.8	(j) 9.5	(k) 19.1	(l) 8.7	(m) 18.7	(n) 0.7	(o) 190	(p) 0.376

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 1	(b) 10.9	(c) 35.2	(d) 21.9	(e) 33.2	(f) 21.9	(g) 31.9	(h) 21.5	(i) 24.8	(j) 30.6	(k) 24.1	(l) 29.5	(m) 23.5	(n) 28.5	(o) 4.8	(p) 60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
23.2	18.0	26.4	22.8	17.6	26.0	22.1	16.8	25.6	75.5	31.0	72.8	29.6	70.3	28.3	27.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 9.0	(b) 8.2	(c) 7.4	(e) -1.8	(f) 39.7	(g) 1.2	(h) 2.0	(i) -2.6	(j) 41.2	(k) -3.3	(l) 42.4	(m) -4.0	(n) 43.5	(o) -4.8	(p) 44.9		
(a) 9.0	(b) 8.2	(c) 7.4	(e) -2.1	(f) 26.4	(g) 1.3	(h) 0.8	(i) -3.0	(j) 26.9	(k) -3.8	(l) 27.4	(m) -4.5	(n) 27.8	(o) -5.5	(p) 28.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	19.8	25.4	24.8	23.5	20.3	16.7	14.3	13.4	14.4	17.9	20.5	22.7	24.3
	DBStd	4.86	2.20	2.13	1.98	2.04	2.37	2.51	2.44	2.52	2.63	2.59	2.58	2.49
	HDD10.0	3	0	0	0	0	0	1	2	1	0	0	0	0
	HDD18.3	517	0	0	0	6	61	123	155	124	39	8	1	0
	CDD10.0	3592	479	416	418	308	209	129	107	138	236	325	382	445
	CDD18.3	1062	221	182	160	64	11	1	1	3	26	74	132	187
	CDH23.3	9526	2008	1554	1219	522	133	16	22	108	416	759	1137	1631
	CDH26.7	3332	808	579	375	95	11	1	1	19	117	257	423	646
Wind	WSAvg	2.9	3.4	3.3	3.0	2.4	2.1	2.3	2.4	2.6	2.9	3.2	3.3	3.3
Precipitation	PrecAvg	859	109	107	80	59	59	41	40	30	31	72	88	120
	PrecMax	1398	635	296	256	363	429	389	214	92	104	320	270	394
	PrecMin	404	0	17	7	2	4	1	1	0	0	7	10	34
	PrecStd	207	98	68	57	75	77	64	44	23	27	54	57	65
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.1	36.2	34.2	31.0	28.0	25.1	25.8	28.8	33.0	35.1	36.2	37.2
		MCWB	23.2	23.5	22.5	18.8	16.9	16.5	14.6	15.7	17.4	18.4	20.0	21.1
	2%	DB	34.5	33.3	31.8	29.1	26.0	23.2	23.5	26.0	29.8	31.8	33.2	34.1
		MCWB	22.9	23.1	22.3	19.2	16.5	15.4	14.1	14.6	16.9	18.3	20.6	22.2
	5%	DB	32.3	31.8	30.1	27.8	24.8	22.1	22.1	24.0	27.6	29.4	31.0	32.0
		MCWB	22.6	22.7	21.4	19.0	16.3	14.9	13.7	14.0	16.4	18.3	20.4	22.0
	10%	DB	30.9	30.1	28.8	26.2	23.5	21.0	20.8	22.3	25.7	27.3	29.0	30.1
		MCWB	22.3	22.1	20.9	18.5	16.1	14.6	13.2	13.4	15.8	18.0	19.8	21.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.6	25.9	24.7	22.5	20.8	18.7	18.7	18.5	20.4	22.5	23.7	25.0
		MCDB	31.6	32.7	29.6	26.3	22.9	21.6	22.4	24.6	27.0	28.8	30.8	32.2
	2%	WB	24.7	24.8	23.7	21.2	19.4	17.6	17.0	17.2	19.3	21.1	22.7	24.1
		MCDB	30.5	30.4	28.1	25.4	23.0	20.8	19.8	22.1	25.2	26.8	29.0	30.3
	5%	WB	24.1	23.9	23.0	20.6	18.4	16.8	15.7	16.1	18.4	20.2	21.9	23.4
		MCDB	29.2	29.0	27.2	24.9	22.2	20.1	19.4	20.7	23.9	25.7	27.5	28.8
	10%	WB	23.5	23.2	22.3	19.8	17.5	15.9	14.7	15.1	17.5	19.4	21.1	22.6
		MCDB	28.3	27.6	26.4	24.2	21.4	19.3	18.7	20.0	22.9	24.7	26.4	27.4
Mean Daily Temperature Range	5% DB	MDBR	10.9	10.5	11.0	12.9	14.4	14.0	15.8	17.1	16.2	14.1	12.3	11.6
		MCDBR	13.6	13.2	13.4	14.7	16.3	15.6	17.7	19.5	19.3	17.7	15.7	14.6
		MCWBR	4.4	4.2	4.6	6.1	7.7	8.3	9.2	9.3	7.9	7.0	5.5	5.0
	5% WB	MCDBR	11.3	11.1	10.5	11.8	11.5	11.5	12.9	15.0	15.2	14.5	12.9	12.2
		MCWBR	4.1	4.0	4.1	5.4	5.9	6.8	7.5	8.0	6.8	6.2	5.0	4.6
Clear-Sky Solar Irradiance	taub	0.390	0.378	0.357	0.338	0.314	0.309	0.299	0.313	0.355	0.378	0.384	0.397	
	taud	2.482	2.524	2.580	2.607	2.627	2.633	2.649	2.585	2.464	2.435	2.457	2.446	
	Ebn at Noon	953	948	938	910	890	873	900	923	924	937	952	948	
	Edh at Noon	117	110	98	87	77	73	74	87	108	118	120	122	
All-Sky Solar Radiation	RadAvg	6.48	5.93	5.12	4.46	3.72	3.08	3.57	4.46	5.47	6.06	6.57	6.53	
	RadStd	0.55	0.53	0.30	0.22	0.21	0.22	0.28	0.34	0.45	0.58	0.61	0.69	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.44	N/A	N/A	N/A	N/A	N/A	-2	-67	+204	+135
(47) Regional (28 neighbors)		+0.41	+0.52	N/A	N/A	N/A	N/A	N/A	-67	+186	+112

Nomenclature: See separate page